

Nonlinear Control of Space Vehicles with Multi-Mass Fuel Slosh Dynamics

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Abstract—This paper studies the maneuvering control problem for space vehicles with fuel slosh dynamics in a zero gravity environment. Multi-mass-spring models are considered for the characterization of the most prominent sloshing modes. The control objective, as is typical for spacecraft maneuvering problems, is to control the translational velocity vector and the attitude of the spacecraft, while attenuating the sloshing modes characterizing the internal dynamics. Subsequently, a nonlinear mathematical model that reflects all of these assumptions is derived. Finally, a Lyapunov-based nonlinear feedback controller is designed to achieve this control objective. A simulation example is included to illustrate the effectiveness of the controller.

Keywords—Nonlinear control, fuel slosh, underactuated system.

I. INTRODUCTION

Ever since the launch of the early high-efficiency rockets, controlling liquid fuel slosh within a launch vehicle has been a major design concern. Moreover, with today's large and complex spacecraft, a substantial mass of fuel is necessary to place them into orbit and to perform orbital maneuvers. The mass of fuel contained in the tanks of a geosynchronous satellite amounts to approximately 40% of its total mass [18]. When the fuel tanks are only partially filled, large quantities of fuel move inside the tanks under translational and rotational accelerations and generate the fuel slosh dynamics.

The traditional treatment of liquid slosh control began with the inclusion of physical barriers, such as baffles and complete compartmentalization, meant to limit the movement of liquid fuel to small amplitudes of high, negligible frequencies. Later, bladders were added to the list of ways to limit these motions. These techniques, although helpful in some cases, do not completely succeed in cancelling the sloshing effects. Moreover, these suppression methods involve adding to the spacecraft structural mass, thereby increasing mission cost. Hence the control system must both assure stability during the thrusting phase and achieve good attitude control while suppressing the slosh dynamics.

There is an extensive body of literature on the interaction of vehicle dynamics and slosh dynamics and their control, but this literature treats only the case of small perturbations to the vehicle dynamics. The control approaches developed for accelerating space vehicles have commonly employed methods of linear control design ([18], [20]) and adaptive

control [1]. A number of related papers following a similar approach are motivated by robotic systems moving liquid filled containers ([6], [8], [9], [19], [21]–[23]). In most of these approaches, suppression of the slosh dynamics inevitably leads to excitation of the transverse vehicle motion through coupling effects; this is a major drawback which has not been adequately addressed in the published literature.

The effect of liquid fuel slosh on spinning spacecraft has also been explored in the literature ([10], [11]). Different slosh motion types - surface waves, bulk fluid motion, and vortices - as well as fluid configurations during spinning are defined [10]. The design of control strategies for a launch vehicle with propellant sloshing has also been studied in several works ([2], [7], [11]). In [2], an advanced linear model of the Saturn V launch vehicle is developed and a linear optimal control law is proposed to control the vehicle. The work in [7] studies the problem of robust control of a launch vehicle subject to aerodynamic, flexible, and slosh mode instabilities. It has been demonstrated that pendulum and mass-spring models can approximate complicated fluid and structural dynamics; such models have formed the basis for many studies on dynamics and control of space vehicles with fuel slosh [13].

As with any continuum model, an infinite number of sloshing modes exists but in this work we will consider finite dimensional discrete mass-spring models. For the sloshing modes, two to three masses at most are sufficient to characterize the sloshing dynamics. Generally, it is more convenient to define the modes by their oscillating frequencies and their assumed damping coefficients. The damping coefficients for the sloshing masses are usually determined by experimental measurements with partially filled tanks [5].

In this paper we will extend our work in [4], [17], where a spacecraft with a partially filled spherical fuel tank was considered and the lowest frequency slosh mode was included in the dynamic model using pendulum and mass-spring analogies. Here we consider multi-mass-spring models for the characterization of the most prominent sloshing modes. As in [4], [17], we assume that a complete set of spacecraft control forces and moments are available to accomplish translational maneuvers. Aerodynamic effects are ignored here, although they can be easily included in the spacecraft dynamics. It is also assumed that the spacecraft

is in a zero gravity environment, but this assumption is for convenience only. These simplifying assumptions render the problem tractable, while still reflecting the important coupling between the slosh dynamics and the rigid body motion of the spacecraft. The control objective, as is typical for spacecraft maneuvering problems, is to control the translational velocity vector and the attitude of the spacecraft, while attenuating the sloshing modes characterizing the internal dynamics. Our main contributions in this paper are (i) the development of a full mathematical model that reflect all of these assumptions and (ii) the design of a Lyapunov-based nonlinear feedback controller. A simulation example is included to illustrate the effectiveness of the controller.

II. MATHEMATICAL MODEL

In this section, we formulate the dynamics of a spacecraft with a single propellant tank and include the prominent fuel slosh modes. We represent the spacecraft as a rigid body (base body) and the sloshing fuel masses as internal bodies, and follow the development in our previous work [3] to express the equations of motion in terms of the spacecraft translational velocity vector, the angular velocity, and the internal (shape) coordinates representing the slosh modes.

To summarize the formulation in [3], let $\mathbf{v} \in \mathbb{R}^3$, $\boldsymbol{\omega} \in \mathbb{R}^3$, and $\boldsymbol{\eta} \in \mathbb{R}^N$ denote the base body translational velocity vector, the base body angular velocity vector, and the vector of internal coordinates, respectively. In these coordinates, the Lagrangian has the form $L = L(\mathbf{v}, \boldsymbol{\omega}, \boldsymbol{\eta}, \dot{\boldsymbol{\eta}})$, which is $SE(3)$ -invariant in the sense that it does not depend on the base body position and attitude. The generalized forces and moments on the spacecraft are assumed to consist of control inputs which can be partitioned into two parts: $\boldsymbol{\tau}_t \in \mathbb{R}^3$ (typically from thrusters) is the vector of generalized control forces that act on the base body and $\boldsymbol{\tau}_r \in \mathbb{R}^3$ (typically from symmetric rotors, reaction wheels, and thrusters) is the vector of generalized control torques that act on the base body. We also assume that the internal dissipative forces are derivable from a Rayleigh dissipation function R . Then, the equations of motion of the spacecraft with internal dynamics are shown to be given by:

$$\frac{d}{dt} \frac{\partial L}{\partial \mathbf{v}} + \hat{\boldsymbol{\omega}} \frac{\partial L}{\partial \mathbf{v}} = \boldsymbol{\tau}_t, \quad (1)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \boldsymbol{\omega}} + \hat{\boldsymbol{\omega}} \frac{\partial L}{\partial \boldsymbol{\omega}} + \hat{\mathbf{v}} \frac{\partial L}{\partial \mathbf{v}} = \boldsymbol{\tau}_r, \quad (2)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{\boldsymbol{\eta}}} - \frac{\partial L}{\partial \boldsymbol{\eta}} + \frac{\partial R}{\partial \dot{\boldsymbol{\eta}}} = 0, \quad (3)$$

where $\hat{\mathbf{a}}$ denotes a 3×3 skew-symmetric matrix formed from $\mathbf{a} = [a_1, a_2, a_3]^T \in \mathbb{R}^3$:

$$\hat{\mathbf{a}} = \begin{bmatrix} 0 & -a_3 & a_2 \\ a_3 & 0 & -a_1 \\ -a_2 & a_1 & 0 \end{bmatrix}.$$

Note that equations (1)-(2) are known as Kirchhoff's equations [12], which can also be expressed in the form of Euler-Poincaré equations. It must be pointed out that in the above formulation it is assumed that no control forces or torques

exist that directly control the internal dynamics. The objective is to simultaneously control the rigid body dynamics and the internal dynamics using only control effectors that act on the rigid body; the control of internal dynamics must be achieved through the system coupling. In this regard, equations (1)-(3) model interesting examples of underactuated mechanical systems. In our previous research [14], [15], we have developed theoretical controllability and stabilizability results for a large class of underactuated mechanical systems using tools from nonlinear control theory. We have also developed effective nonlinear control design methodologies [16] that we applied to several examples of underactuated mechanical systems, including underactuated space vehicles [4], [17].

We now derive a multi-mass-spring model of the sloshing fuel where the oscillation frequencies of the mass-spring elements represent the prominent sloshing modes [18].

Consider a rigid spacecraft moving on a plane as indicated in Figure 1, where v_x, v_z are the axial and transverse components, respectively, of the velocity of the center of the fuel tank, and θ denotes the attitude angle of the spacecraft with respect to a fixed reference. The fluid is modeled by moment of inertia I_0 assigned to a rigidly attached mass m_0 and point masses m_i , $i = 1, \dots, N$, whose relative positions along the spacecraft fixed z -axis is denoted by s_i . The locations $h_0 > 0$ and $h_i > 0$ are referenced to the center of the tank. A restoring force $-k_i s_i$ acts on the mass m_i whenever the mass is displaced from its neutral position $s_i = 0$. A thrust F is produced by a gimbaled thrust engine as shown in Fig. 1, where δ denotes the gimbal deflection angle, which is considered as one of the control inputs. A pitching moment M is also available for control purposes. The constants in the problem are the spacecraft mass m and moment of inertia I ; the fuel masses m_0, m_i ; the distance b between the body z -axis and the spacecraft center of mass location along the longitudinal axis, and the distance d from the gimbal pivot to the spacecraft center of mass. If the tank center is in front of the spacecraft center of mass then $b > 0$. The parameters m_i, h_i, k_i and b depend on the shape of the fuel tank, the characteristics of the fuel and the fill ratio of the fuel tank. To preserve the static properties of the liquid, the sum of all the masses must be the same as the fuel mass m_f , and the center of mass of the model must be at the same elevation as that of the fuel, i.e.

$$m_0 + \sum_{i=1}^N m_i = m_f,$$

$$m_0 h_0 = \sum_{i=1}^N m_i h_i.$$

Let $\hat{i}$ and $\hat{k}$ be the unit vectors along the spacecraft-fixed longitudinal and transverse axes, respectively, and denote by (x, z) the inertial position of the center of the fuel tank. The position vector of the center of mass of the vehicle can then be expressed in the spacecraft-fixed coordinate frame as

$$\vec{r} = (x - b)\hat{i} + z\hat{k}.$$

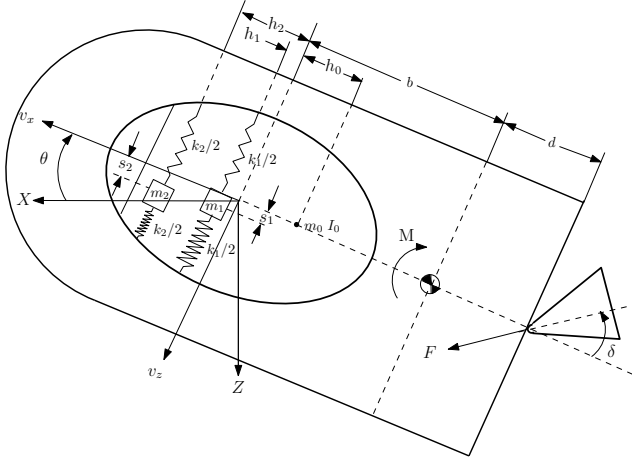


Figure 1. A multiple slosh mass-spring model for a spacecraft.

Clearly, the inertial velocity of the vehicle can be computed as

$$\dot{\vec{r}} = v_x \hat{i} + (v_z + b\dot{\theta}) \hat{k},$$

where we have used the fact that $v_x = \dot{x} + z\dot{\theta}$ and $v_z = \dot{z} - x\dot{\theta}$.

Similarly, the position vectors of the fuel masses $m_0, m_i, \forall i$, in the spacecraft-fixed coordinate frame are given, respectively, by

$$\begin{aligned} \vec{r}_0 &= (x - h_0) \hat{i} + z \hat{k}, \\ \vec{r}_i &= (x + h_i) \hat{i} + (z + s_i) \hat{k}, \quad \forall i. \end{aligned}$$

Assuming h_i are constants, the inertial velocities can be computed as

$$\begin{aligned} \dot{\vec{r}}_0 &= v_x \hat{i} + (v_z + h_0 \dot{\theta}) \hat{k}, \\ \dot{\vec{r}}_i &= (v_x + s_i \dot{\theta}) \hat{i} + (v_z - h_i \dot{\theta} + \dot{s}_i) \hat{k}, \quad \forall i. \end{aligned}$$

The total kinetic energy can now be expressed as

$$T = \frac{1}{2} m \dot{\vec{r}}^2 + \frac{1}{2} m \dot{\vec{r}}_0^2 + \frac{1}{2} \sum_{i=1}^N m_i \dot{\vec{r}}_i^2 + \frac{1}{2} (I + I_0) \dot{\theta}^2.$$

Since gravitational effects are ignored, there is no gravitational potential energy. The elastic potential energy is given by

$$U = \frac{1}{2} \sum_{i=1}^N k_i s_i^2.$$

Thus, under the indicated assumptions, the Lagrangian ($L = T - U$) can be computed as

$$\begin{aligned} L &= \frac{1}{2} m [v_x^2 + (v_z + b\dot{\theta})^2] + \frac{1}{2} m_0 [v_x^2 + (v_z + h_0 \dot{\theta})^2] \\ &+ \frac{1}{2} \sum_{i=1}^N m_i [(v_x + s_i \dot{\theta})^2 + (v_z - h_i \dot{\theta} + \dot{s}_i)^2] \\ &+ \frac{1}{2} (I + I_0) \dot{\theta}^2 - \frac{1}{2} \sum_{i=1}^N k_i s_i^2. \end{aligned}$$

We include dissipative effects due to fuel slosh, described by damping constants c_i . A fraction of kinetic energy of sloshing fuel is dissipated during each cycle of the motion. When the damping is small, it can be represented accurately by equivalent linear viscous damping. Even with baffles, the damping ratio is seldom greater than about 0.05. We will include the damping via a Rayleigh dissipation function R given by

$$R = \frac{1}{2} \sum_{i=1}^N c_i \dot{s}_i^2.$$

Applying equations (1)-(3) with

$$\boldsymbol{\eta} = \begin{pmatrix} s_1 \\ \vdots \\ s_N \end{pmatrix}, \quad \mathbf{v} = \begin{pmatrix} v_x \\ 0 \\ v_z \end{pmatrix}, \quad \boldsymbol{\omega} = \begin{pmatrix} 0 \\ \dot{\theta} \\ 0 \end{pmatrix},$$

$$\boldsymbol{\tau}_t = \begin{pmatrix} F \cos \delta \\ 0 \\ F \sin \delta \end{pmatrix}, \quad \boldsymbol{\tau}_r = \begin{pmatrix} 0 \\ M + F(b + d) \sin \delta \\ 0 \end{pmatrix},$$

the equations of motion can be obtained as

$$(m + m_f)(\dot{v}_x + \dot{\theta} v_z) + mb\dot{\theta}^2 + \sum_{i=1}^N m_i (s_i \ddot{\theta} + 2\dot{s}_i \dot{\theta}) = F \cos \delta, \quad (4)$$

$$(m + m_f)(\dot{v}_z - \dot{\theta} v_x) + mb\ddot{\theta} + \sum_{i=1}^N m_i (\ddot{s}_i - s_i \dot{\theta}^2) = F \sin \delta, \quad (5)$$

$$\begin{aligned} \hat{I} \ddot{\theta} + \sum_{i=1}^N m_i (s_i (\dot{v}_x + \dot{\theta} v_z) - h_i \ddot{s}_i + 2s_i \dot{s}_i \dot{\theta}) \\ + mb(\dot{v}_z - \dot{\theta} v_x) = M + F(b + d) \sin \delta, \end{aligned} \quad (6)$$

$$m_i (\ddot{s}_i + \dot{v}_z - \dot{\theta} v_x - h_i \ddot{\theta} - s_i \dot{\theta}^2) + k_i s_i + c_i \dot{s}_i = 0, \quad \forall i, \quad (7)$$

where

$$\hat{I} = I + I_0 + mb^2 + m_0 h_0^2 + \sum_{i=1}^N m_i (h_i^2 + s_i^2).$$

The control objective is to design feedback controllers so that the controlled spacecraft accomplishes a given planar maneuver, that is a change in the translational velocity vector and the attitude of the spacecraft, while suppressing the fuel slosh modes.

III. FEEDBACK CONTROL LAW

Consider the model of a spacecraft with a gimballed thrust engine shown in Figure 1. If the thrust F during the fuel burn is a positive constant, and if the gimbal deflection angle and pitching moment are zero, $\delta = M = 0$, then the spacecraft and fuel slosh dynamics have a relative equilibrium defined by

$$v_z = \bar{v}_z, \theta = \bar{\theta}, \dot{\theta} = 0, s_i = 0, \dot{s}_i = 0, \forall i,$$

where $\bar{v}_z$ and $\bar{\theta}$ are arbitrary constants. Without loss of generality in our subsequent analysis, we consider the relative equilibrium at the origin, i.e. $\bar{v}_z = 0, \bar{\theta} = 0$. Note that the relative equilibrium corresponds to the vehicle axial velocity

$$v_x(t) = \frac{F}{m + m_f}t + v_{x0}, \quad t \leq t_b,$$

where v_{x0} is the initial axial velocity of the spacecraft and t_b is the fuel burn time.

Now assume the axial acceleration term a_x is not significantly affected by small gimbal deflections, pitch changes and fuel motion (an assumption verified in simulations). Consequently, equation (4) becomes:

$$\dot{v}_x + \dot{\theta}v_z = \frac{F}{m + m_f}. \quad (8)$$

Substituting this approximation leads to the following equations of motion for the transverse, pitch and slosh dynamics:

$$(m + m_f)(\dot{v}_z - \dot{\theta}v_x(t)) + m\ddot{\theta} + \sum_{i=1}^N m_i(\ddot{s}_i - s_i\dot{\theta}^2) = F \sin \delta, \quad (9)$$

$$\begin{aligned} \hat{I}\ddot{\theta} + \sum_{i=1}^N m_i \left[s_i \left(\frac{F}{m + m_f} \right) - h_i \ddot{s}_i + 2s_i \dot{s}_i \dot{\theta} \right] \\ + mb(\dot{v}_z - \dot{\theta}v_x) = M + F(b + d) \sin \delta, \end{aligned} \quad (10)$$

$$\ddot{s}_i + \dot{v}_z - \dot{\theta}v_x(t) - h_i \ddot{\theta} - s_i \dot{\theta}^2 + \frac{k_i}{m_i} s_i + \frac{c_i}{m_i} \dot{s}_i = 0, \quad \forall i, \quad (11)$$

where $v_x(t)$ is considered as an exogenous input. Our subsequent analysis is based on the above equations of motion for the transverse, pitch and slosh dynamics of the vehicle.

We now design a nonlinear controller to stabilize the relative equilibrium at the origin of the equations (9)-(11). Our control design is based on a Lyapunov function approach. By defining control transformations from (δ, M) to new control inputs (u_1, u_2) , the equations (9)-(11) can be written as:

$$\dot{v}_z = u_1 + \dot{\theta}v_x(t), \quad (12)$$

$$\ddot{\theta} = u_2, \quad (13)$$

$$\ddot{s}_i = -\omega_i^2 s_i - 2\zeta_i \omega_i \dot{s}_i - u_1 + h_i u_2 + s_i \dot{\theta}^2, \quad \forall i, \quad (14)$$

where

$$\omega_i^2 = \frac{k_i}{m_i}, \quad 2\zeta_i \omega_i = \frac{c_i}{m_i}, \quad \forall i.$$

Here ω_i and $\zeta_i, \forall i$, denote the undamped natural frequencies and damping ratios, respectively.

Now, we consider the following candidate Lyapunov function for the system (12)-(14):

$$V = \frac{r_1}{2}v_z^2 + \frac{r_2}{2}\theta^2 + \frac{r_3}{2}\dot{\theta}^2 + \frac{r_4}{2} \sum_{i=1}^N (\dot{s}_i^2 + \omega_i^2 s_i^2 - 2h_i \dot{s}_i \dot{\theta}),$$

where r_1, r_2, r_3 , and r_4 are positive constants. We assume that

$$\mu = r_3 - r_4 \sum_{i=1}^N h_i^2 > 0$$

so that the function V is positive definite.

Remark: Let $\mathbf{z} = (\dot{\theta}, \dot{s}_1, \dots, \dot{s}_N)^T$ and let $\mathbf{Q} \in \mathbb{R}^{(N+1) \times (N+1)}$ denote the symmetric matrix corresponding to the quadratic form

$$\mathbf{z}^T \mathbf{Q} \mathbf{z} = r_3 \dot{\theta}^2 + r_4 \sum_{i=1}^N (\dot{s}_i^2 - 2h_i \dot{s}_i \dot{\theta}).$$

Clearly, this quadratic form is positive definite if and only if the leading principal minors of the matrix $\mathbf{Q}$ are all positive. It is easy to show that the condition holds if $\mu > 0$.

The time derivative of V along the trajectories of (12)-(14) is

$$\begin{aligned} \dot{V} = -2r_4 \sum_{i=1}^N \zeta_i \omega_i \dot{s}_i^2 + [r_1 v_z - r_4 \sum_{i=1}^N (\dot{s}_i - h_i \dot{\theta})] u_1 \\ + [r_1 v_x(t) v_z + r_2 \dot{\theta} + \mu u_2 \\ + r_4 \sum_{i=1}^N (h_i \omega_i^2 s_i + 2\zeta_i \omega_i h_i \dot{s}_i + s_i \dot{s}_i \dot{\theta} - h_i s_i \dot{\theta}^2)] \dot{\theta}. \end{aligned}$$

Clearly, the feedback laws

$$u_1 = -l_1 [r_1 v_z - r_4 \sum_{i=1}^N (\dot{s}_i - h_i \dot{\theta})], \quad (15)$$

$$\begin{aligned} u_2 = -\frac{1}{\mu} [r_2 \dot{\theta} + l_2 \dot{\theta} + r_1 v_x(t) v_z \\ + r_4 \sum_{i=1}^N (h_i \omega_i^2 s_i + 2\zeta_i \omega_i h_i \dot{s}_i + s_i \dot{s}_i \dot{\theta} - h_i s_i \dot{\theta}^2)], \end{aligned} \quad (16)$$

where l_1 and l_2 are positive constants, yield

$$\dot{V} = -l_1 [r_1 v_z - r_4 \sum_{i=1}^N (\dot{s}_i - h_i \dot{\theta})]^2 - l_2 \dot{\theta}^2 - 2r_4 \sum_{i=1}^N \zeta_i \omega_i \dot{s}_i^2,$$

which satisfies $\dot{V} \leq 0$. Using LaSalle's principle, it is easy to prove asymptotic stability of the origin of the closed loop defined by the equations (12)-(14) and the feedback control laws (15)-(16). Note that the positive gains $l_j, j = 1, 2$, can be chosen arbitrarily to achieve good closed loop responses.

IV. SIMULATION

The feedback control laws developed in the previous sections are implemented here for a spacecraft. The first two slosh modes are included to demonstrate the effectiveness of the control laws. The physical parameters used in the simulations are given in Table 1.

Table I
PHYSICAL PARAMETERS.

Parameter	Value	Parameter	Value
m	400 kg	F	825 N
I	500 kg · m ²	I_0	7.5 kg · m ²
m_0	100 kg	k_1	72 N/m
m_1	40 kg	k_2	90 N/m
m_2	10 kg	c_1	4.5 N · s/m
h_0	0.1 m	c_2	2.4 N · s/m
h_1	0.15 m	b	0.6 m
h_2	0.4 m	d	0.4 m

We consider stabilization of the spacecraft in orbital transfer, suppressing the transverse and pitching motion of the spacecraft and sloshing of fuel while the spacecraft is accelerating. In other words, the control objective is to stabilize the relative equilibrium corresponding to a constant axial spacecraft acceleration of 1.5 m/s^2 and $v_z = \theta = \dot{\theta} = s_i = \dot{s}_i = 0$, $i = 1, 2$.

In this section, we demonstrate the effectiveness of the Lyapunov-based controller (15)-(16) by applying to the complete nonlinear system (4)-(7).

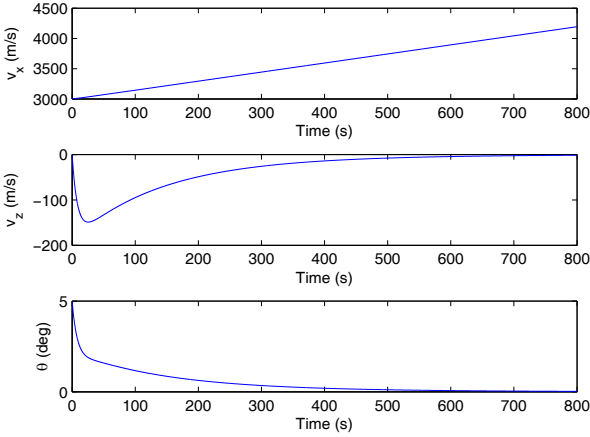


Figure 2. Time responses of v_x , v_z and θ .

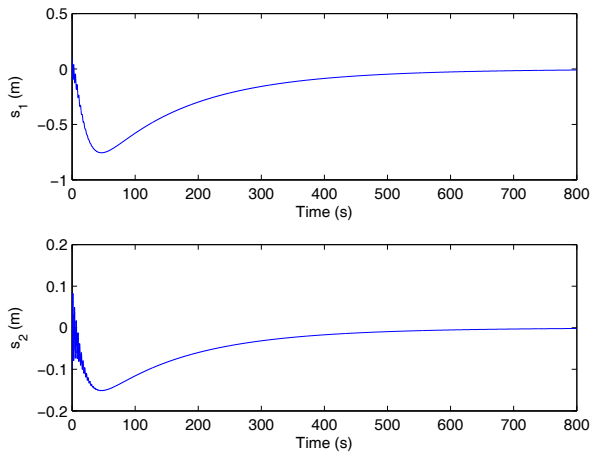


Figure 3. Time responses of s_1 and s_2 .

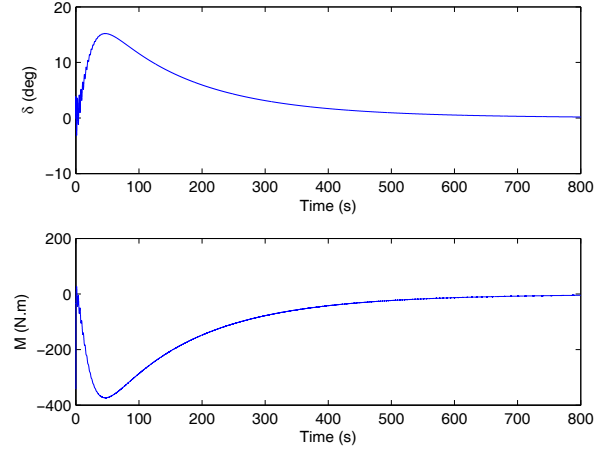


Figure 4. Gimbal deflection angle δ and pitching moment M .

Time responses shown in Figures 2-4 correspond to the initial conditions $v_{x0} = 3000 \text{ m/s}$, $v_{z0} = 0$, $\theta_0 = 5^\circ$, $\dot{\theta}_0 = 0$, $s_{10} = 0.1 \text{ m}$, $s_{20} = -0.1 \text{ m}$, and $\dot{s}_{10} = \dot{s}_{20} = 0$. We assume a fuel burn time of 800 s. As can be seen in the figures, the transverse velocity, attitude angle, and the slosh states converge to the relative equilibrium at zero while the axial velocity v_x increases and $\dot{v}_x$ tends asymptotically to 1.5 m/s^2 . Note that there is a trade-off between good responses for the directly actuated degrees of freedom (the transverse and pitch dynamics) and good responses for the unactuated degree of freedom (the slosh dynamics); the controller given by (15)-(16) with parameters $r_1 = 5 \times 10^{-6}$, $r_2 = 80$, $r_3 = 10$, $r_4 = 0.01$, $l_1 = 2000$, $l_2 = 1000$ represents one example of this balance.

V. CONCLUSIONS

We have developed a complete nonlinear dynamical model for a spacecraft with multiple slosh modes. We have designed a Lyapunov-based nonlinear feedback control law that achieves stabilization of the pitch and transverse dynamics as well as suppression of the slosh modes, while the spacecraft accelerates in the axial direction. The effectiveness of this control feedback law has been illustrated through a simulation example. The many avenues considered for future research include problems involving multiple propellant tanks and three dimensional maneuvers. Future research also includes designing nonlinear control laws that achieve robustness, insensitivity to system and control parameters, and improved disturbance rejection.

VI. ACKNOWLEDGMENTS

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REFERENCES

- [1] J. M. Adler, M. S. Lee, and J. D. Saugen, "Adaptive Control of Propellant Slosh for Launch Vehicles," *SPIE Sensors and Sensor Integration*, Vol. 1480, 1991, pp. 11-22.
- [2] T. R. Blackburn and D. R. Vaughan, "Application of Linear Optimal Control and Filtering Theory to the Saturn V Launch Vehicle," *IEEE Transactions on Automatic Control*, Vol. 16, No. 6, 1971, pp. 799-806.

- [3] S. Cho, N. H. McClamroch, and M. Reyhanoglu, "Dynamics of Multi-body Vehicles and Their Formulation as Nonlinear Control Systems," *Proceedings of American Control Conference*, 2000, pp. 3908-3912.
- [4] S. Cho, N. H. McClamroch, and M. Reyhanoglu, "Feedback Control of a Space Vehicle with Unactuated Fuel Slosh Dynamics," *Proceedings of AIAA Guidance, Navigation, and Control Conference*, AIAA 2000-4046, Vol. 1, pp. 354-359.
- [5] F.T. Dodge, The new "Dynamic Behavior of Liquids in Moving Containers," 2000, Southwest Research Institute, San Antonio, TX.
- [6] J. T. Feddema, C. R. Dohrmann, G. G. Parker, R. D. Robinett, V. J. Romero, and D. J. Schmitt, "Control for Slosh-Free Motion of an Open Container," *IEEE Control Systems Magazine*, 1997, pp. 29-36.
- [7] J. Freudenberg and B. Morton, "Robust Control of a Booster Vehicle Using H^∞ and SSV Techniques," *Proceedings of the 31st IEEE Conference on Decision and Control*, 1992, pp. 2448-2453.
- [8] M. Grundelius, "Iterative Optimal Control of Liquid Slosh in an Industrial Packaging Machine," *Proceedings of the 39th IEEE Conference on Decision and Control*, 2000, pp. 3427-3432.
- [9] M. Grundelius and B. Bernhardsson, "Control of Liquid Slosh in an Industrial Packaging Machine," *Proceedings of IEEE International Conference on Control Applications*, 1999, pp. 1654-1659.
- [10] C. Hubert, "Behavior of Spinning Space Vehicles with Onboard Liquids," Hubert Astronautics Technical Report B3007, NASA/KSC Contract NAS10-02016, 2003.
- [11] C. Hubert, "Design and Flight Performance of a System for Rapid Attitude Maneuvers by a Spinning Vehicle," *Proceedings of the 27th Annual AAS Guidance and Control Conference*, AAS 04-078, 2004.
- [12] L. Meirovitch and M. K. Kwak, "State Equations for a Spacecraft with Maneuvering Flexible Appendages in Terms of Quasi-Coordinates," *Applied Mechanics Reviews*, Vol. 42, No. 11, 1989, pp. 161-170.
- [13] L. D. Peterson, E. F. Crawley, and R. J. Hansman, "Nonlinear Fluid Slosh Coupled to the Dynamics of a Spacecraft," *AIAA Journal*, Vol. 27, No. 9, 1989, pp. 1230-1240.
- [14] M. Reyhanoglu, A.J. van der Schaft, N.H. McClamroch, and I. Kolmanovsky, "Nonlinear Control of a Class of Underactuated Systems," *Proceedings of IEEE Conference on Decision and Control*, 1996, pp. 1682-1687.
- [15] M. Reyhanoglu, A.J. van der Schaft, N.H. McClamroch, and I. Kolmanovsky, "Dynamics and Control of a Class of Underactuated Mechanical Systems," *IEEE Transactions on Automatic Control*, Vol. 44, No. 9, 1999, pp. 1663-1671.
- [16] M. Reyhanoglu, S. Cho, and N.H. McClamroch, "Discontinuous Feedback Control of a Special Class of Underactuated Mechanical Systems," *International Journal of Robust and Nonlinear Control*, Vol. 10, No. 4, 2000, pp. 265-281.
- [17] M. Reyhanoglu, "Maneuvering Control Problems for a Spacecraft with Unactuated Fuel Slosh Dynamics," *Proceedings of IEEE Conference on Control Applications*, 2003, pp. 695-699.
- [18] M. J. Sidi, *Spacecraft Dynamics and Control*, Cambridge Aerospace Series, Cambridge University Press, 1997.
- [19] K. Terashima and G. Schmidt, "Motion Control of a Cart-Based Container Considering Suppression of Liquid Oscillations," *Proceedings of IEEE International Symposium on Industrial Electronics*, 1994, pp. 275-280.
- [20] B. Wie, *Space Vehicle Dynamics and Control*, AIAA Education Series, 1998.
- [21] K. Yano, T. Toda, and K. Terashima "Sloshing Suppression Control of Automatic Pouring Robot by Hybrid Shape Approach," *Proceedings of IEEE Conference on Decision and Control*, 2001, pp. 1328-1333.
- [22] K. Yano, S. Higashikawa, and K. Terashima, "Liquid Container Transfer Control on 3D Transfer Path by Hybrid Shaped Approach," *Proceedings of IEEE International Conference on Control Applications*, 2001, pp. 1168-1173.
- [23] K. Yano and K. Terashima, "Robust Liquid Container Transfer Control for Complete Sloshing Suppression," *IEEE Transactions on Control Systems Technology*, Vol. 9, No. 3, 2001, pp. 483-493.